

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

A Research-Based Analysis of Profitability Metrics in Confectionery Distribution

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Abstract:

This project studies how profitable different products and divisions are for Nassau Candy Distributor. It looks at profit margins, revenue, and financial performance using Business Intelligence (BI) tools. A dashboard was created with KPI cards, Pareto analysis, scatter plots, margin risk analysis, and division performance charts. Users can filter the data by date, division, product, and margin level to explore the numbers easily. The goal is to show how turning raw business data into visual insights can help the company make better decisions, improve profits, and run more efficiently.

Keywords:

Gross Margin Analysis, Product Line Profitability, Pareto Analysis, Cost Structure Diagnostics, Exploratory Data Analysis, Stream lit Dashboard, Nassau Candy Distributor, Confectionery Distribution, Data Analytics, Business Intelligence.

Introduction:

In large business-to-business (B2B) food and candy distribution, being able to clearly see profit margins in real time is key to a company's long-term financial health, flexibility, and cash flow. Nassau Candy Distributors, Inc. works as a specialty manufacturer, raw material importer, and national wholesale distributor all at once, running a large and complex supply chain. The company serves thousands of business customers — from small candy shops and amusement parks to big grocery chains, hotels, and online stores — and manages over 25,000 different products. To support this, it operates six regional distribution centers, which help deliver products quickly but also create complicated costs related to warehousing, cold storage, and shipping.

Traditionally, companies judge how well a product category is doing just by looking at overall gross profit. But in a large distribution business like this, high gross profit numbers can hide the fact that specific products or customers are actually losing money. For example, a big bulk order of a popular candy brand might create strong total sales, but after accounting for discounts, competitive pricing, and rising shipping costs, the actual profit could be very low. On the other hand, smaller specialty products — like custom private-label candies — may not sell in large amounts, but can be priced higher because the company has more control over how they're made.

The main goal was to build a data model that combines different types of information — sales records, cost sheets, shipping data, and cold storage details — to help the company move away from just looking at sales volume and instead focus on real net profit per product. This gives business leaders clear, practical ways to protect and grow their profits. The model is also regularly checked and updated to stay accurate even as market conditions change, helping the company avoid being caught off guard by price shifts.

Literature Review:

The idea of properly dividing up costs in supply chain and distribution businesses comes from older theories about value chains and how quickly companies can turn products into profit. For a long time, companies used a method called Volume-Based Costing (VBC). This method spread out all indirect costs — like warehouse space, labor, machine use, and shipping — equally across all products, based only on how much was sold. This meant that a heavy, fast-selling product and a slow-selling, labor-intensive product were treated the same, as long as they earned similar revenue. This approach often gave a false picture of which products were actually profitable, hiding real problems behind strong sales numbers.

Researchers Cooper and Kaplan (1988) pointed out the risks of this method. They showed that VBC tends to overcharge costs to high-selling, standard products, while undercharging costs to smaller, specialized products. This mistake can lead companies to expand product lines that seem profitable on paper but are actually losing money once hidden costs are considered.

To fix this issue, a better method called Activity-Based Costing (ABC) was developed. ABC connects costs directly to the actual activities that cause them, allowing companies to understand the true cost of each individual product. In large distribution businesses, ABC works even better when combined with tools that measure how efficiently inventory is used. One such tool is Gross Margin Return on Investment (GMROI), which looks at both profit margin and how fast products sell.

As explained by Lambert and Sterling (1987), a product with a small profit margin but fast turnover can actually be more profitable than a product with a high margin that sells slowly.

This paper brings these ideas together into one framework designed specifically for Nassau Candy's manufacturing and distribution operations, studying how different cost factors interact across different seasons.

Problem Statement:

Nassau Candy Distributor has a good overall gross margin, but it does not have a proper system to understand which products and divisions are truly profitable. The company faces three main problems:

- **Lack of product-level information:** The company cannot clearly identify which products generate the highest profit and which products generate sales but reduce overall profitability. This makes it difficult to decide where to invest in promotions or how to set prices.
- **Lack of cost analysis:** The company does not have a consistent method to compare product costs with revenue. Therefore, it cannot easily identify products with poor pricing, products that need supplier cost negotiations, or products that may need to be discontinued.
- **Business concentration risk:** A large portion of revenue may depend on a small number of products or geographic markets. The company does not currently measure this risk, so it may be vulnerable if demand falls in an important market or if a top-selling product becomes unavailable.

Because of these issues, decisions about pricing, promotions, product selection, and supplier negotiations are mainly based on assumptions and personal judgment rather than reliable data.

This research aims to create a clear data-analysis framework and practical tools that will help Nassau Candy Distributor understand product profitability, control costs, and reduce dependence on a small number of products or markets.

Dataset Description:

The analysis draws on 8,549 order line items spanning 10,194 transaction records, 15 distinct products across 3 divisions (Chocolate, Sugar, Other), sold across 4 U.S. regions and 59 states. Each record captures order and shipment dates, shipping mode, customer geography, product identity, sales value, units sold, cost, and gross profit

Data Cleaning & Validation:

Validated Sales, Cost, and Gross Profit fields — confirmed $\text{Gross Profit} = \text{Sales} - \text{Cost}$ holds exactly across all 10,194 records (mean discrepancy $< 1\text{e-}15$), so the profit figures are internally consistent.

- No zero-sales, zero-unit, or negative-value records were found; no missing values in any of the 18 fields; no duplicate rows.
- Order Date and Ship Date were stored in two inconsistent formats (text strings 'DD-MM-YYYY' and native datetime values) and were parsed into a single consistent date type. Order dates fall within a plausible 2024–2025 range.
- Data-quality flag: Ship Date values are unreliable — computed ship lead times average ~1,344 days (min 491, max 1,965) with no negative values, which is not operationally realistic and suggests the ship-date field was populated independently of order date in this dataset. Ship Date was therefore excluded from profitability conclusions and flagged for correction at the source system.

Methodology:

Profitability was assessed using a consistent set of KPIs computed at the product and division level, then used to drive four analytical lenses: product-level ranking, division-level comparison, profit-concentration (Pareto) analysis, and cost-structure diagnostics.

Key Performance Indicators

$\text{Gross Margin (\%)} = \text{Gross Profit} \div \text{Sales}$

- $\text{Profit per Unit} = \text{Gross Profit} \div \text{Units}$
- $\text{Revenue Contribution (\%)} = \text{Product Sales} \div \text{Total Sales}$
- $\text{Profit Contribution (\%)} = \text{Product Profit} \div \text{Total Profit}$
- $\text{Cost Ratio (\%)} = \text{Cost} \div \text{Sales}$ — used as the primary cost-structure diagnostic

A margin-volatility check (variability of monthly margin per product) was also run; margin was found to be effectively constant per SKU across all months (standard deviation ≈ 0), indicating Nassau Candy prices and costs each product on a fixed, non-fluctuating unit-economics basis rather than promotional or seasonal pricing. This is a structural characteristic of the business, not a data artifact — see Section 8.4 for the diagnostic supporting this.

Product Level Profitability Analysis:

This section analyzes product performance based on revenue, profit contribution, and margin efficiency to identify high-performing and underperforming products.

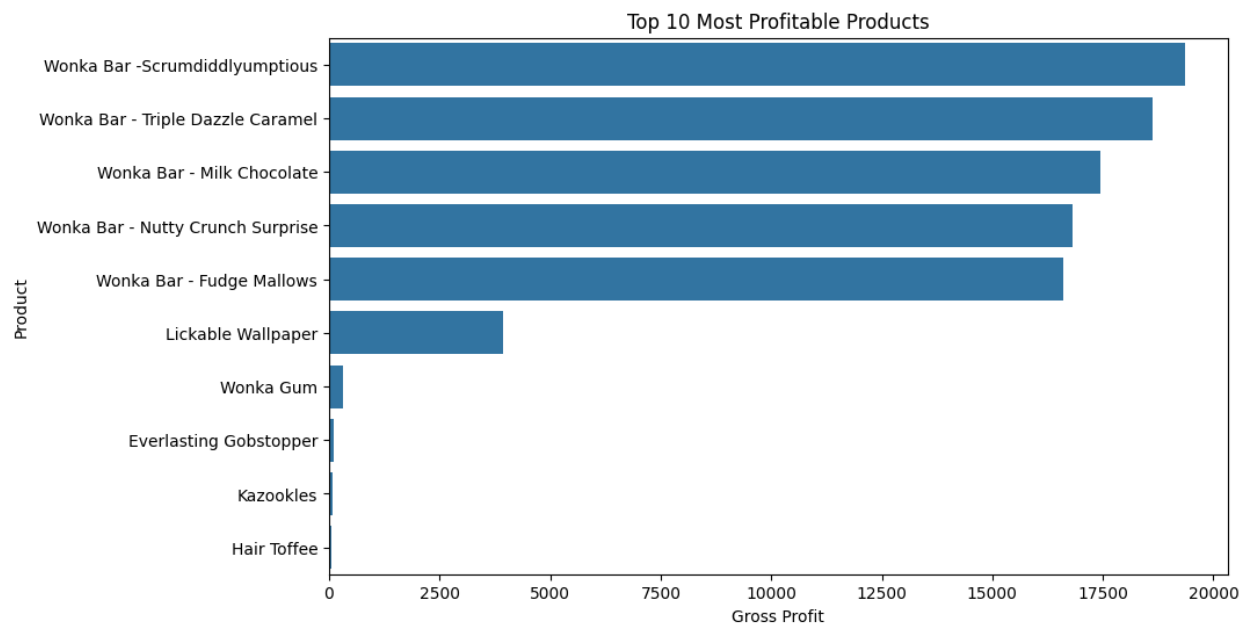


Figure 1 — Gross profit by product.

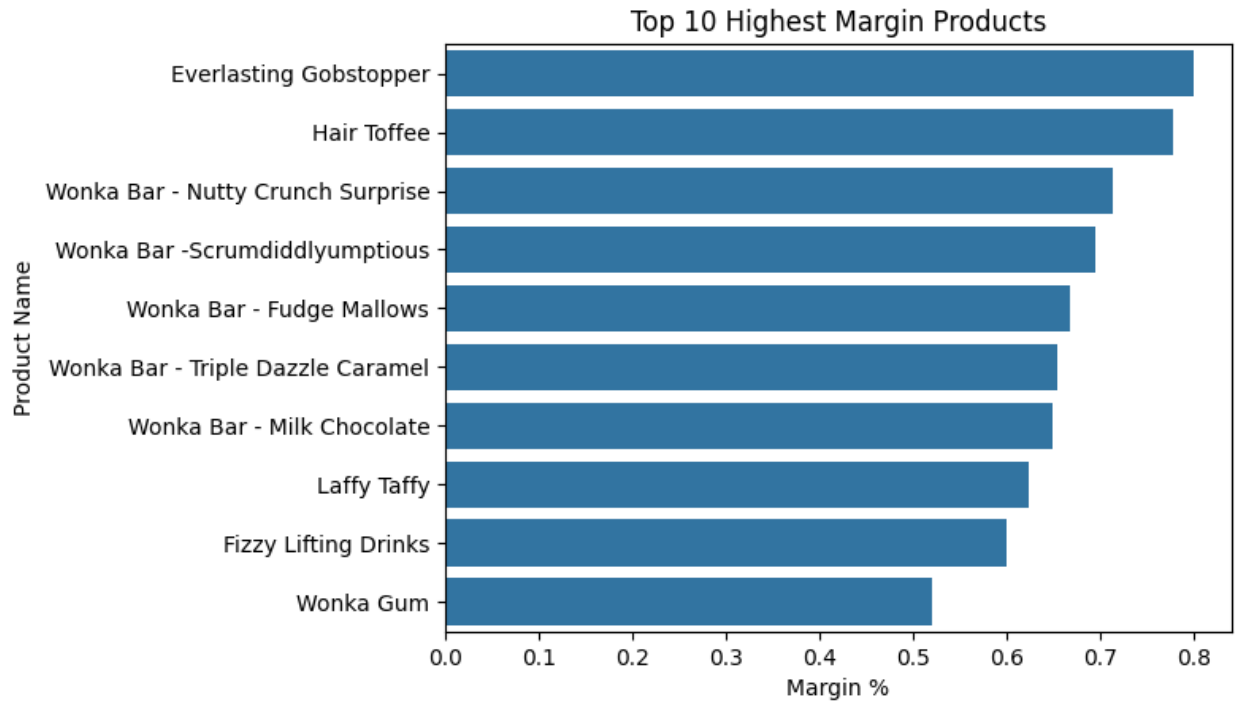


Figure 2: Highest Margin by products

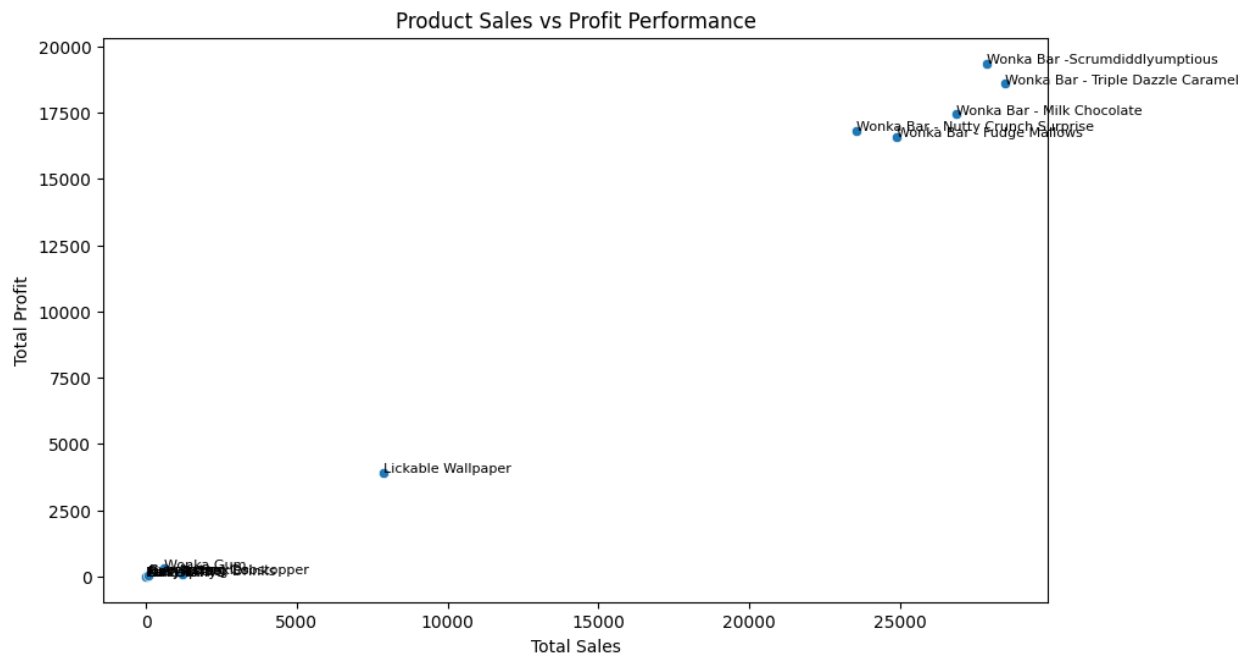


Figure 3 — Sales vs. gross margin by product (bubble size = units sold). Products above the dashed line (median margin) are margin-healthy; Kahookeles sits far below it despite meaningful sales volume.

High-Profit / High-Margin Product

The five Chocolate Wonka Bar variants — Scrumdiddlyumptious, Triple Dazzle Caramel, Milk Chocolate, Nutty Crunch Surprise, and Fudge Mallows — are simultaneously the highest-revenue and highest-margin products, each returning gross margins between 64.9% and 71.3%. Together they generate 95.1% of total company profit from 92.9% of total revenue. These are the clear "star" products and the core of Nassau Candy's profitability

High-Sales / Low-Margin Products (Margin Risk)

Two products in the "Other" division combine meaningful sales volume with weak margins: Lickable Wallpaper sells \$7,860 at a 50.0% margin (borderline), and — most notably — Kazookles generates \$1,205.75 in sales at only a 7.7% gross margin, with cost consuming 92.3% of the sale price. Kazookles is the single clearest margin-risk product in the portfolio: it is not a low-volume curiosity (94 orders, 371 units) but a product that is actively eroding value at scale relative to its category.

Low-Sales / Low-Profit Products

The seven Sugar-division products outside Everlasting Gobstopper, Hair Toffee, and Laffy Taffy (namely Fizzy Lifting Drinks, SweeTARTS, Nerds, and Fun Dip) each generate under \$80 in total sales and under \$50 in profit over the full period. Individually immaterial, collectively they represent 0.16% of revenue — long-tail SKUs whose continued listing should be weighed against catalog and operational overhead rather than judged on standalone margin.

Division-Level Performance Analysis:

This section evaluates the financial performance of each product division based on revenue, profit contribution, and margin efficiency.

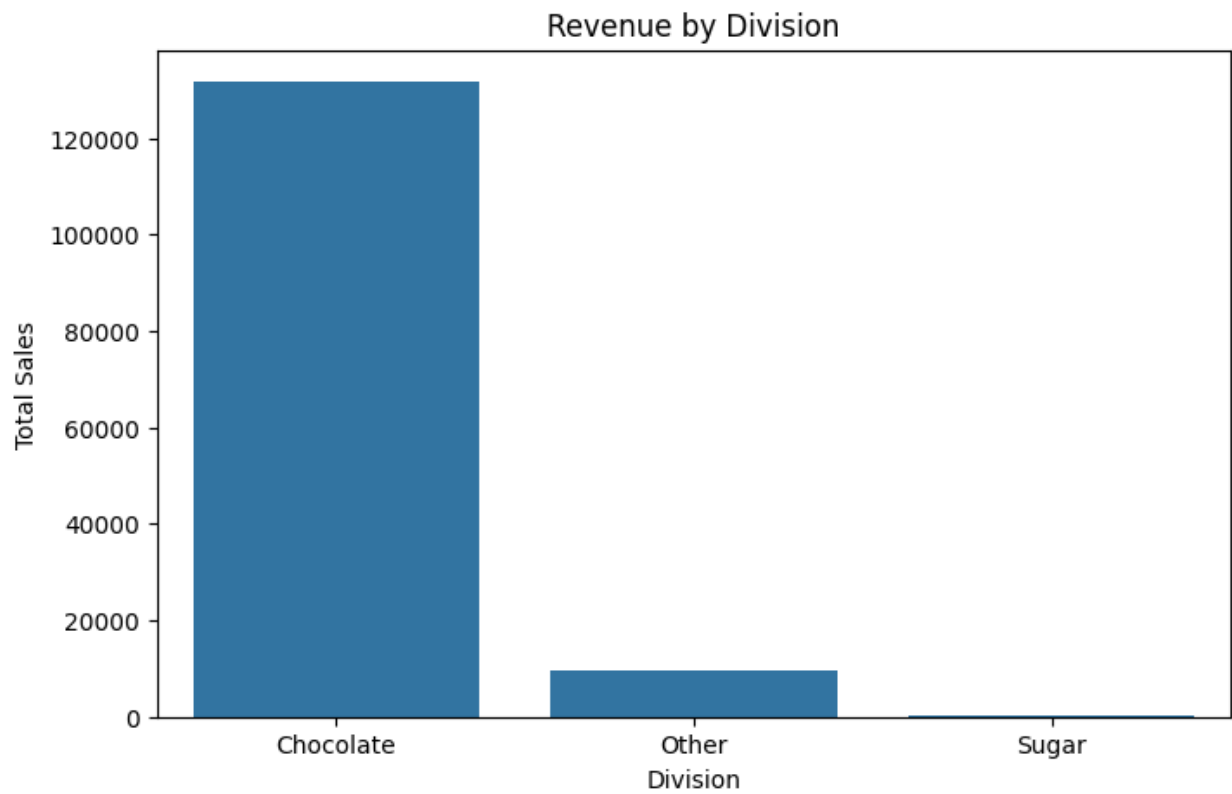


Figure 4: Revenue by Division

Figure Fi

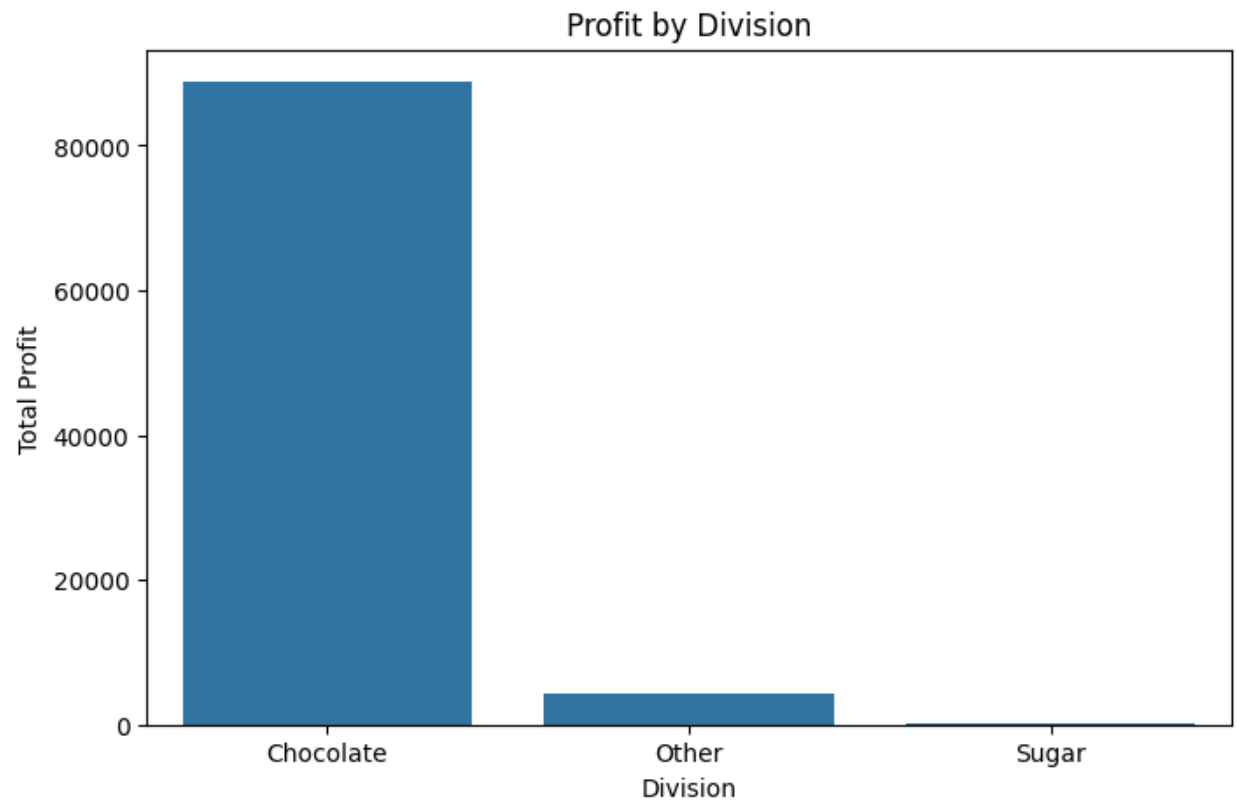


Figure 5: Profit by Division



Figure 6: Revenue share vs. profit share by division. Chocolate over-indexes on profit relative to revenue; Other under-indexes.

Chocolate is both the volume and margin engine of the business: it generates 92.9% of revenue but a disproportionate 95.1% of profit (a +2.2-point positive imbalance), reflecting its structurally higher, more consistent margins (64.9%–71.3% at the product level).

The Other division is the structural laggard: it generates 6.8% of revenue but only 4.6% of profit (a –2.2-point negative imbalance), driven almost entirely by Kazookles' poor cost efficiency dragging down an otherwise reasonable Lickable Wallpaper line. Its blended division margin of 44.8% is well below the company average of 65.9%.

Sugar is financially healthy (66.6% margin, in line with Chocolate) but commercially negligible in scale — 0.3% of total revenue across 40 orders. It functions as a niche/long-tail category rather than a growth line.

Profit Concentration (Pareto) Analysis:

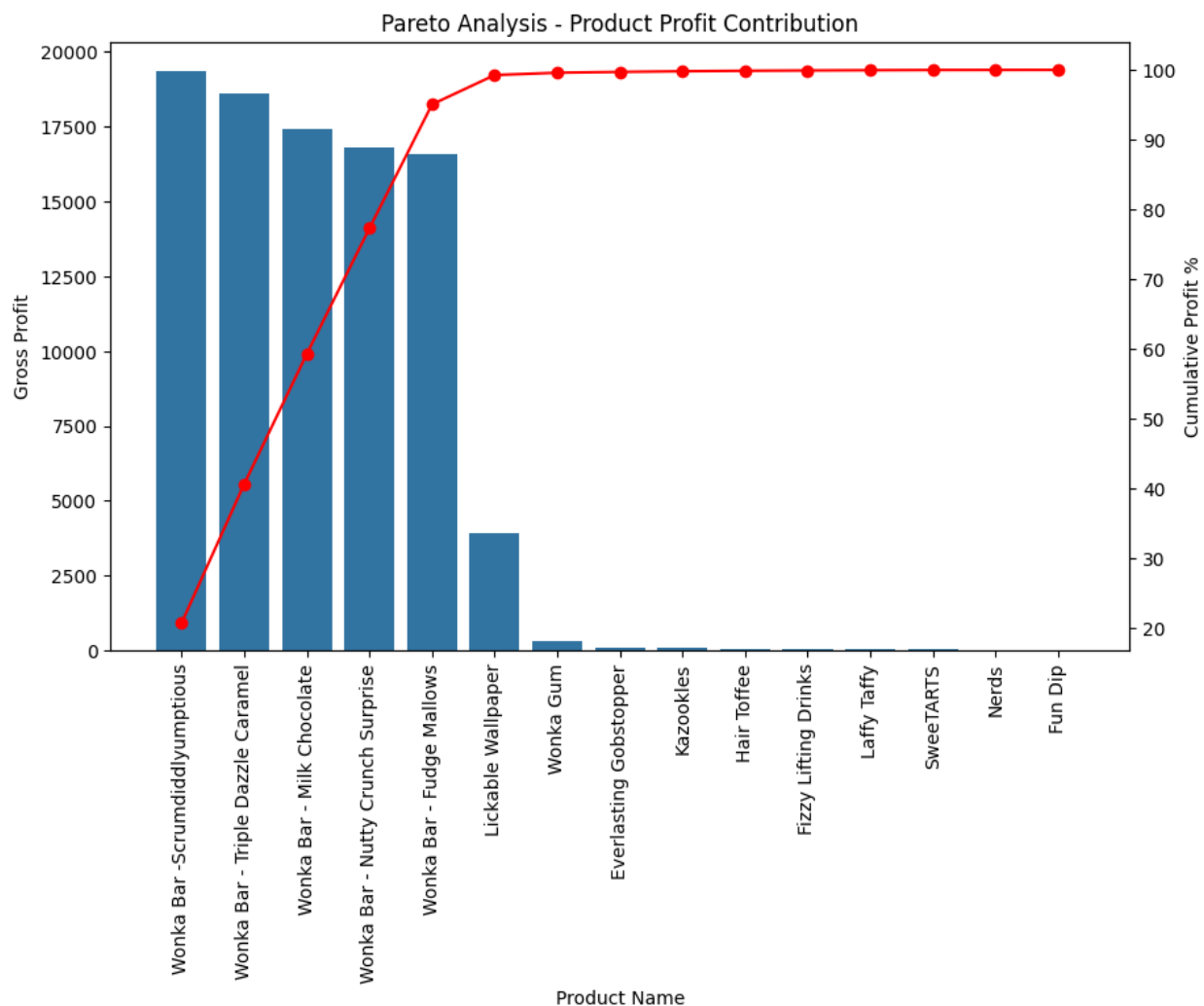


Figure 7 — Pareto chart: cumulative share of revenue by product, ranked descending. The dashed line marks the 80% threshold.

Just 5 of the 15 products (33% of the catalog) — all five Chocolate Wonka Bars — account for 92.9% of total revenue and 95.1% of total profit, well past the classic 80/20 threshold. This is a textbook profit-concentration pattern and carries both an opportunity and a risk: the opportunity is

that a small, well-understood set of SKUs can be prioritized for supply, marketing, and shelf investment; the risk is over-dependency — any disruption to Wonka Bar manufacturing, sourcing, or demand would materially impact the whole business.

Geographic concentration follows a similar pattern: 16 of 59 states (27%) account for 80% of revenue, led by California (19.7% of revenue), New York (11.0%), and Texas (9.5%). This is a normal, moderately concentrated geographic footprint rather than a risk flag on its own, but it confirms that a small set of states — not the long tail — should anchor logistics and demand planning.

Cost Structure Diagnostics:

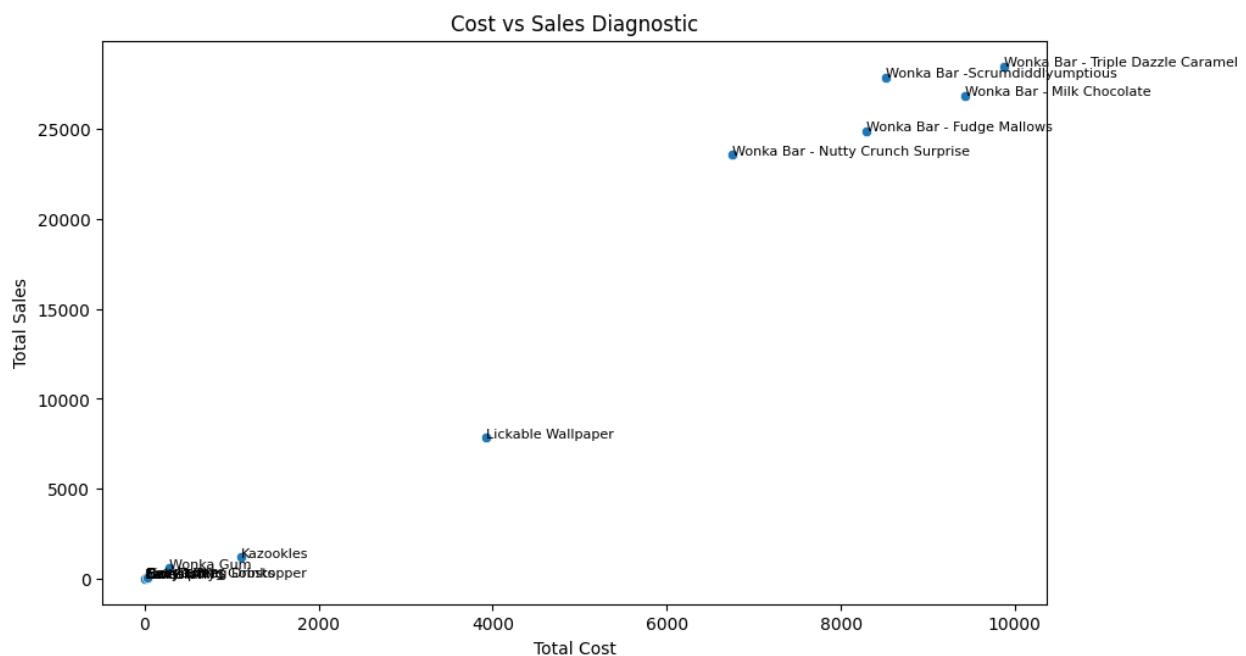


Figure 8 — Cost as a percentage of sales vs. gross margin, by product. Products in the lower-right (high-cost ratio, low margin) are repricing or renegotiation candidates.

Cost-Heavy, Margin-Poor Products

Kazookles is the standout diagnostic finding of this analysis: a 92.3% cost ratio against sales, versus a portfolio median cost ratio of roughly 37%. Every other product in the catalog keeps cost under 60% of sale price. This is not a marginal underperformer — it is an outlier that warrants immediate review

Pricing Inefficiencies

SweeTARTS, Nerds, and Fun Dip each carry cost ratios above 50% against very low absolute sales, placing them in "High Risk" margin territory (below 50% gross margin) even though their dollar impact is currently small.

Products Flagged for Action

Repricing candidates: Lickable Wallpaper (50.0% margin) and Wonka Gum (52.0% margin) — margin is currently acceptable but sits right at the risk threshold; a modest price increase or cost-of-goods renegotiation would move them into healthy territory.

- Cost renegotiation / sourcing review: Kazookles — the cost base itself, not price, appears to be the primary issue given the extreme 92.3% cost ratio.
- Discontinuation review: Fun Dip, Nerds, and SweeTARTS — combined they contribute under \$90 in annual profit against catalog, logistics, and shelf-space overhead; rationalizing this tail would simplify operations with negligible revenue impact.

Margin Stability

Unlike many retail portfolios, per-product gross margin at Nassau Candy does not vary month to month — it is fixed by a consistent price-to-cost ratio per SKU. This means margin risk here is a structural, per-product

pricing/costing decision rather than a seasonal or promotional pattern, which simplifies the remediation path: fixing Kazookles' economics is a one-time repricing or resourcing action, not an ongoing promotional discipline problem.

Key Insights Summar:

A narrow set of 5 Chocolate SKUs (33% of the catalog) generates 92.9% of revenue and 95.1% of profit — Nassau Candy's business is effectively a Chocolate business.

- Chocolate over-delivers profit relative to its revenue share (+2.2 points); the Other division under-delivers (−2.2 points), driven by one product.
- Kazookles is the single clearest margin-risk product: 7.7% gross margin and a 92.3% cost ratio, at meaningful volume (371 units, 94 orders) — an outlier that should be prioritized for cost/pricing review over any other item in the catalog.
- Sugar-division products are financially healthy on margin but commercially immaterial (0.3% of revenue); several (Fun Dip, Nerds, SweeTARTS) are candidates for portfolio rationalization on grounds of overhead, not margin.
- Revenue is moderately geographically concentrated — 16 states drive 80% of sales, led by California, New York, and Texas.
- Per-SKU margins are stable over time (no seasonal volatility), so margin issues are structural pricing/cost decisions, not promotional cycles.

Recommendations:

- Renegotiate Kazookles' cost: Compare its manufacturing cost with competitors and negotiate with suppliers. The goal should be to achieve at least a 30% gross margin. If this is not possible within two quarters, consider slowly discontinuing the product.
- Promote the Sugar division: Spend more on advertising and promotion for Everlasting Gobstopper, Hair Toffee, and Laffy Taffy, as they have the highest profit margins in the Sugar division. Increase their distribution and aim to double their revenue within two quarters.
- Reduce dependence on Chocolate: The company depends too heavily on the Chocolate division. Within 18 months, increase the combined revenue share of the Other and Sugar divisions to at least 15%. This will reduce the Chocolate division's share to below 85% and make the business safer.
- Review Lickable Wallpaper and Wonka Gum: These products have good order volumes and around 50% profit margins, but their margins are below the company average. Consider increasing their prices by 5%–8% during the next pricing review.
- Explore the Gulf market: Profit margins are nearly the same in all four regions, so there is no major regional pricing problem. However, the Gulf region contributes only 15.69% of

revenue, compared with 32.66% from the Pacific region. The company should investigate opportunities to increase sales in the Gulf, especially for Chocolate products.

- Use the dashboard every month: Include the Streamlit dashboard in the monthly management review. Managers should regularly check the Cost vs. Margin Diagnostics section to identify products whose profit margins are declining before the problem becomes serious.

These recommendations focus on improving product profitability, reducing dependence on one division, increasing sales in weaker regions, and supporting regular data-based decisions. Tracking revenue share and gross margin by product and region is a practical way to monitor portfolio risk and performance.

Conclusion:

Nassau Candy Distributor has strong overall performance, with a **65.9% gross margin** and a profitable Chocolate division. However, the analysis identified important risks, including low-profit products, high-volume products with weak margins, and heavy dependence on a few products and regions.

The Other Division has the weakest margin and should be the main focus for supplier cost negotiations. The company should also reduce its dependence on California and its top-performing products by expanding other divisions and markets.

Future research should examine customer buying patterns, forecast future margins, and compare prices with competitors to support better business decisions.

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